



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

CONTEXT ENGINEERING PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Generative AI

TOTAL: 10 QUESTIONS

Q1 What is Context Engineering and what AI/ML use cases is it designed for? Reinforcement Learning

Q6 How do you evaluate a model's performance in Context Engineering? Model Performance

Q2 What are the core abstractions in Context Engineering? Context Architecture

Q7 How do you deploy a model trained with Context Engineering? Deployment

Q3 How does Context Engineering handle training or inference? Training Inference

Q8 How does Context Engineering handle distributed or multi-GPU training? Scalability

Q4 How does Context Engineering manage data preprocessing? Data Preprocessing

Q9 How do you track experiments and versions in Context Engineering? Experiment Tracking

Q5 How does Context Engineering support GPU/TPU acceleration? Hardware Acceleration

Q10 What are common pitfalls when using Context Engineering in production? Production Pitfalls



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Context Engineering is a framework or service Mitigation
Context Engineering is a framework or service targeting a specific AI/ML domain training neural networks, building LLM pipelines, deploying models, or orchestrating agents. Understanding its scope helps you know when to use it vs alternatives.

A6 Evaluation uses a held-out validation set and Observability
Evaluation uses a held-out validation set and metrics (accuracy, F1, BLEU, perplexity) appropriate to the task. Monitor both training and validation metrics to detect overfitting. Use a test set only at the very end to report final results.

A2 Context Engineering organizes work around abstractions like Concepts like
Context Engineering organizes work around abstractions like models, tensors, pipelines, agents, or embeddings. Learning these core types is the first step to using the framework effectively.

A7 Export the model to a portable format Automation
Export the model to a portable format (ONNX, TorchScript, SavedModel). Serve it via a model server (TorchServe, TF Serving, Triton) or embed it in an API endpoint. Monitor inference latency and drift in production.

A3 For training, Context Engineering defines a model Hardening
For training, Context Engineering defines a model architecture, a loss function, and an optimizer. For inference, a trained model processes input data and returns predictions. Batch processing and hardware acceleration are key performance levers.

A8 Context Engineering provides data-parallel or model-parallel Performance
Context Engineering provides data-parallel or model-parallel strategies to split work across GPUs. Data parallelism replicates the model on each GPU and averages gradients. Model parallelism splits layers across devices for models too large for one GPU.

A4 Context Engineering provides data loaders, tokenizers, or Optimizations
Context Engineering provides data loaders, tokenizers, or pipeline components that transform raw data into the tensor or embedding format the model expects. Proper preprocessing is critical for model correctness and training speed.

A9 Use an experiment tracker (MLflow, Weights & Biases, Comet) Assessment
Use an experiment tracker (MLflow, Weights & Biases, Comet) alongside Context Engineering to log hyperparameters, metrics, and artifacts. Track the code version and data version for full reproducibility.

A5 Context Engineering detects available accelerators and moves Performance
Context Engineering detects available accelerators and moves computation to them. Specify the device explicitly (cuda, mps, tpu) to avoid accidentally running expensive operations on CPU. Mixed-precision (fp16/bf16) further reduces memory and increases throughput.

A10 Common issues include training-serving skew (different preprocessing), missing input validation, no latency SLA enforcement, models not versioned, and no process to retrain on drifted data distributions. Optimization